

Types of Logical Data Models

A **logical data model** describes how data is logically organized and how different pieces of data are related to one another.

It provides a structure for representing data **without focusing on exactly how the data is physically stored on a computer.**

A logical data model helps answer questions such as:

- How is the data organized?
- How are different records related?
- What rules govern those relationships?
- How can users or applications navigate or retrieve the data?

Major Logical Data Models

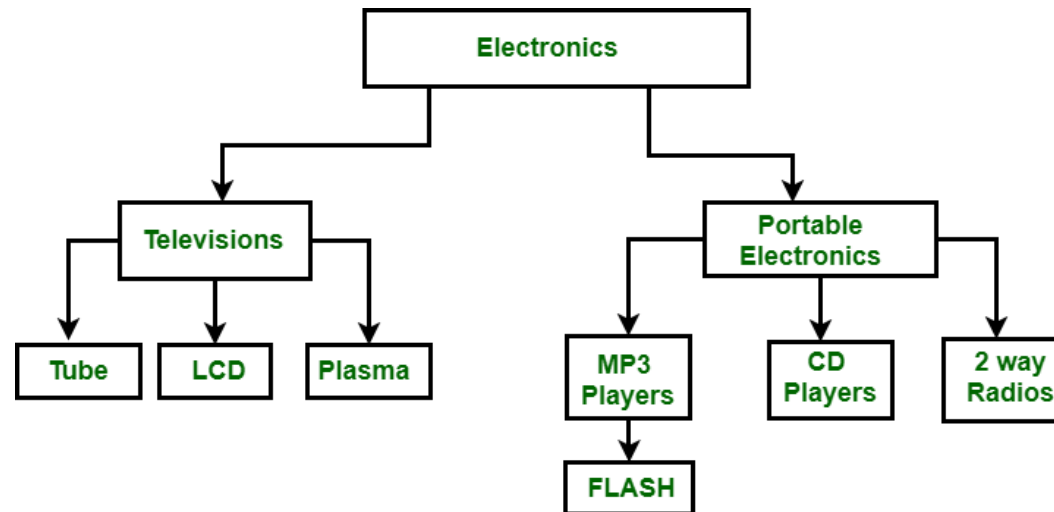
Historically, three important logical data models are the **hierarchical**, **network**, and **relational** models.

Logical Data Model	Organization	Basic Structure
Hierarchical	Parent-child relationships	Tree
Network	Records connected through links	Network/graph
Relational	Data organized into relations	Tables

1. Hierarchical Model

The hierarchical [data model](#) is the oldest type of the data model. It was developed by IBM in 1968. It organizes data in a tree-like structure. Hierarchical model consists of the following :

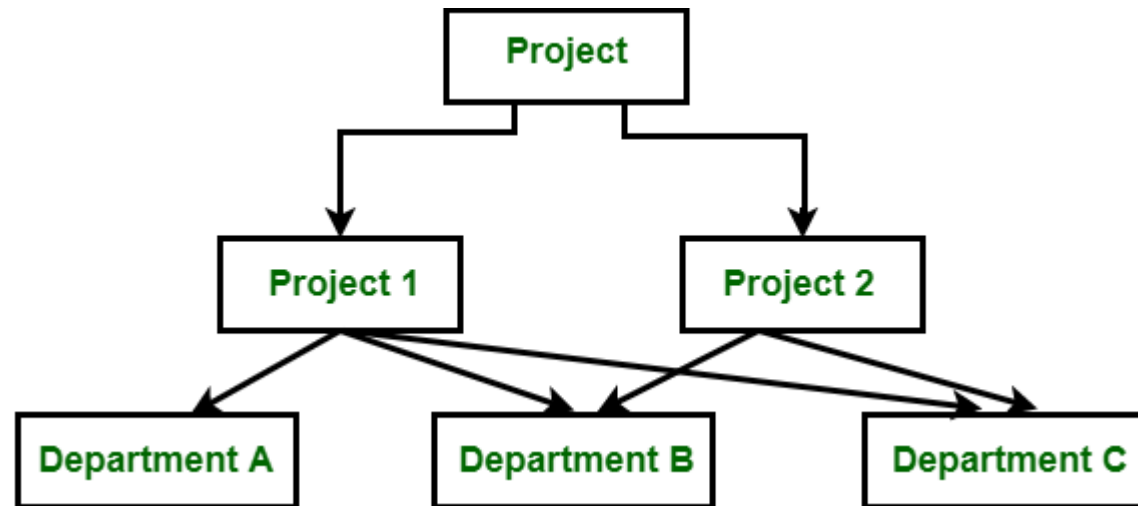
- It contains nodes which are connected by branches.
- The topmost node is called the root node.
- If there are multiple nodes appear at the top level, then these can be called root segments.
- Each node has exactly one parent.
- One parent may have many children.



2. Network Model

The **network model** is an early database model in which data is organized as **records connected to other records through links or relationships**. It was developed before the relational model became dominant.

Unlike the **hierarchical model**, which organizes data as a tree where each child normally has only one parent, the network model allows a record to be connected to **multiple parent records**. This makes it better suited for representing more complex relationships.



3. Relational Data Model

The **relational data model** is a logical data model that organizes data into **relations**, commonly represented as **tables**. Each table represents a particular type of information, and relationships between tables are established primarily through **keys**.

The relational model was introduced by **Edgar F. Codd** in 1970 and became the foundation for most modern relational database management systems.

EMPLOYEE (Relation / Table)				
Primary Key ↓				Foreign Key (references DEPARTMENT) ↓
EmpID (PK)	FirstName	LastName	HireDate	DepartmentID (FK)
1001	Emily	Carter	2019-01-15	10
1002	Michael	Johnson	2020-03-20	20
1003	Sophia	Martinez	2021-07-01	30
1004	Daniel	Lee	2022-02-10	20
1005	Olivia	Brown	2023-06-12	10

DEPARTMENT (Relation / Table)	
Primary Key ↓	
DepartmentID (PK)	DepartmentName
10	Accounting
20	Information Technology
30	Human Resources
40	Marketing